# Mohammad Seraj

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## Professional Summary

AI/ML Engineer with 1.5 years of experience designing and deploying NLP, Generative AI, and deep learning solutions. Built production RAG systems, LLM fine-tuning pipelines, and scalable AI services using LangChain, PyTorch, FastAPI, and AWS across healthcare and enterprise domains.

## Work Experience

**AI Solutions Engineer** June 2026 – Present

*MGT (Amerisource Solutions Pvt. Ltd.)* *Remote*

- **Project: Agentic AI and Intelligent Document Processing**
- Developing enterprise-grade **agentic AI workflows** using LangChain and LangGraph to orchestrate LLM-powered tasks, tool execution, contextual reasoning, and multi-step information processing.
- Building **OCR-based document intelligence pipelines** that combine text extraction with AI models for document understanding, contextual analysis, and structured information extraction from complex and scanned documents.
- Implementing AI-powered document analysis and automated data extraction workflows using **LLM reasoning** and prompt engineering to improve information retrieval, classification, and response accuracy.
- Developing scalable backend services and REST APIs using **FastAPI**, integrating AI models, document-processing pipelines, and application components into modular, maintainable services.
- Integrating **LiteLLM** for LLM observability, model usage tracking, request monitoring, and performance analysis across AI-powered workflows.
- Contributing to application containerization and cloud deployment using **Docker**, **Azure**, Git-based collaboration, and CI/CD pipelines to support reliable development and deployment workflows.
- *Stack: Python, Agentic AI, LangChain, LangGraph, FastAPI, Large Language Models (LLMs), LLM Reasoning, Prompt Engineering, OCR, Document Intelligence, LiteLLM Observability, REST APIs, Docker, Azure, Git, CI/CD*

**Freelance AI Engineer / Independent Research** Nov 2025 – June 2026

*Self-Employed* *Remote*

- Co-authored manuscript (ct-2026-00508r) – *Quantum-Physics Inspired ML Model for Partition Coefficient Prediction* submitted to **Journal of Chemical Theory and Computation** (under review).

**AI Developer** Apr 2025 – Oct 2025

*Anervea Data Labs Private Limited* *Pune, Maharashtra, India*

- **Project: Clinical Trial Patient Recruitment Platform (GenAI / RAG System)**
- Architected and deployed an AI-powered system that reduced patient identification time from **several months to under 30 minutes**, automating end-to-end eligibility screening from EHR and claims data at scale.
- Engineered end-to-end **NLP ingestion pipelines** using AWS Textract and AWS Bedrock (Claude Sonnet 3.5) to extract and structure clinical trial protocol components trial titles, inclusion/exclusion criteria into standardized, machine-readable formats, enabling automated protocol parsing across **multiple concurrent trials**.
- Designed and implemented a **RAG architecture with PGVector**, enabling semantic retrieval of ICD-10, CPT, and LOINC medical codes alongside demographic and gender-specific eligibility criteria for precise patient matching, reducing manual review effort by **˜70%**.
- Integrated with **Komodo Workspace** (HIPAA-compliant healthcare dataset) to operationalize cohort generation surfacing each matched patient's most recent HCOs and HCPs, enabling researchers to directly contact and recruit eligible volunteers without exposing protected patient data.
- Implemented **LLM evaluation and monitoring pipelines** for RAG workflows using **RAGAS metrics** (faithfulness, answer relevance, context precision) and **LangSmith tracing** to analyze retrieval quality, prompt behavior, and end-to-end response reliability.
- Deployed **HIPAA-compliant microservices** on AWS (EC2, S3) and Snowflake, enabling near real-time trial feasibility analysis and accelerating time-to-market for novel therapies.
- **Project: BioBERT Fine-Tuning for Clinical NLP Classification**
- Fine-tuned **BioBERT** for clinical-trial NLP classification using PEFT/QLoRA with 4-bit quantization via bitsandbytes, achieving efficient low-memory training and improved domain-adapted inference performance.
- Containerized all NLP and GenAI services with **Docker**; managed CI/CD workflows via Jenkins and GitHub, ensuring reproducible and scalable cloud deployments.
- *Stack: Python, LangChain, LangGraph, FastAPI, AWS (EC2, S3, Textract, Bedrock), Vector Databases (PGVector), RAG Pipelines, LLM Evaluation (RAGAS), LangSmith Monitoring, Snowflake, Docker, Jenkins, BioBERT, Hugging Face Transformers, PEFT/QLoRA, bitsandbytes, PyTorch, CUDA*

**Research Assistant** Jun 2024 – Apr 2025

*NeuroCare-eLab* *New Delhi, India*

- Led computational drug discovery research on **LogP prediction and molecular solubility estimation**, combining classical and quantum descriptors (QM, QMPSA, $\Delta$G) to engineer highly predictive molecular feature sets.
- Benchmarked GNN, XGBoost, Random Forest, and ExtraTrees models with 5-fold cross-validation; achieved state-of-the-art LogP performance with **GNN ($R^2$=0.959, MAE=0.115, MSE=0.222)**, outperforming XGBoost ($R^2$=0.932) and feature-augmented ExtraTrees ($R^2$=0.849).
- Designed end-to-end computational pipelines covering data preprocessing, model training, evaluation, and documentation, enabling reproducible experimentation that accelerated drug discovery workflows.
- *Stack: Python, PyTorch, scikit-learn, GNNs, RDKit, Modred, DeepChem, Feature Engineering (Classical + Quantum Descriptors)*

**Researcher — M.Tech Thesis** Jun 2023 – May 2024

*Indraprastha Institute of Information Technology Delhi (IIIT-D)* *New Delhi, India*

- Engineered a **volumetric 3D deep learning model** on `.nii` MRI files, improving Alzheimer's diagnostic accuracy by **64% over 2D baselines**; Thesis Link.
- Developed multi-view YOLOv8n classifiers across axial, coronal, and sagittal planes surpassing VGG16 baselines (coronal 80.5%, sagittal 78.1%, axial 71.0%), and built an ensemble achieving **75% overall accuracy**.
- *Stack: PyTorch, TensorFlow, YOLOv8n, OpenCV*

## Projects

**Autonomous Voice-Enabled Dealership Agent** View Code

*CrewAI, LangChain, Llama 3.3, FastAPI, Deepgram, Docker*

- Built a multi-agent orchestration system (CrewAI + Llama 3.3) automating customer-service workflows: test-drive scheduling, intent detection, and knowledge-base retrieval — **replacing manual handling of repetitive customer queries end-to-end**.
- Engineered a low-latency voice pipeline with Deepgram Nova-2 (STT) and Aura (TTS); implemented tool-level guardrails and strict prompt controls that **eliminated date/time booking hallucinations** to achieve near-zero error rate in scheduling logic.
- Implemented an async FastAPI backend managing concurrent audio streams, session chat history, and dynamic knowledge retrieval at production scale, supporting **multiple simultaneous voice sessions** without degradation.

**GenAI Document Intelligence Platform** View Code

*Python, LangChain, Hugging Face, FastAPI, Qdrant, Docker*

- Designed an end-to-end RAG pipeline for intelligent document understanding, semantic search, and QA over enterprise document corpora.
- Containerized inference services via FastAPI and Docker; iterative prompt optimization improved answer accuracy and reduced hallucination rate in production.

## Education

**Indraprastha Institute of Information Technology, Delhi (IIIT-D)** 2022 – 2024

*M.Tech in Computational Biology* — CGPA: 7.96 / 10.0 *New Delhi, India*

- *Relevant Coursework: Natural Language Processing, Machine Learning in Biomedical Applications, Big Data Mining in Healthcare, Biomedical Image Processing*

**Hindustan College of Science and Technology, Mathura** 2018 – 2022

*B.Tech in Biotechnology* — CGPA: 7.80 / 10.0 *Mathura, UP, India*

## Technical Skills

**Core Strengths:** RAG Systems, Large Language Models, Agentic AI, LLM Fine-Tuning, NLP, Production ML Pipelines

**Generative AI / LLMs:** LangChain, LangGraph, CrewAI, Prompt Engineering, Tool Calling, PEFT (QLoRA), 4-bit Quantization, Embeddings, AWS Bedrock

**NLP:** BioBERT, Hugging Face Transformers, NER, Text Classification, Tokenization, spaCy, NLTK, Document Understanding

**ML / Deep Learning:** Neural Networks, GNN, XGBoost, Random Forest, SVM, Cross-Validation, Feature Engineering, Hyperparameter Tuning

**Cloud & MLOps:** AWS (EC2, S3, Textract, Bedrock), Docker, FastAPI, CI/CD, Jenkins, GitHub Actions, MLflow, Model Serving

**Databases & Vector Stores:** PGVector, Qdrant, FAISS, Snowflake, SQL

**Frameworks & Libraries:** PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, Streamlit, OpenCV, RDKit, DeepChem

**Languages & Tools:** Python, SQL, Git, GitHub, VS Code, Google Colab, CUDA, cuDNN

## Achievements

**DBT GAT-B (2022) — All India Rank 133**

*Department of Biotechnology, Government of India*

- Cleared the national biotechnology aptitude examination conducted by the Department of Biotechnology, Government of India.